```
import numpy as np
import pandas as pd

data = {
    "Customer ID": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Age Group": [
        "20-25", "30-35", "25-30", "20-25", "35+", "35+",
        "25+", "20-25", "25-30", "30-35"
    ],
    "Rating(1-5)": [4, 5, 6, -1, 3, 3, 4, 7, 2, 5],
    "Hotel": [
        "Taj", "Taj", "Lemon Tree", "Red Fox", "Red Fox",
        "Taj", "Taj", "Lemon Tree", "Lemon Tree", "Taj"
    ],
    "Food preference": [
        "Veg", "Non-veg", "Veg", "Non-veg", "Veg",
        "Non-veg", "Vegetarian", "Non-veg", "Veg", "Vegetarian"
    ],
    "Bill": [
        300, 200, 1322, 1234, 989, 1905, 1000, 2999, 3456, 3456
    ],
    "No of pax": [2, 3, 2, 2, 2, 1, 1, 3, 3, 5],
    "Estimated salary": [
        45000, 50000, 120000, 45000, 122220, 211234,
        345623, 30000, 99999, 99999
    ]
}

df = pd.DataFrame(data)

df = df.drop_duplicates()

df["Hotel"] = df["Hotel"].replace(["Taj"], "Taj")

df["Food preference"] = df["Food preference"].replace(
    {"Vegetarian": "Veg"}
)

df["Rating(1-5)"] = pd.to_numeric(
    df["Rating(1-5)"], errors="coerce"
)

df.loc[
    (df["Rating(1-5)"] < 1) | (df["Rating(1-5)"] > 5),
    "Rating(1-5)"
] = np.nan

df["Bill"] = pd.to_numeric(df["Bill"], errors="coerce")
df.loc[df["Bill"] <= 0, "Bill"] = np.nan
```

```
df["Estimated salary"] = pd.to_numeric(
    df["Estimated salary"], errors="coerce"
)

df.loc[df["Estimated salary"] <= 0, "Estimated salary"] = np.nan

df["Age Group"] = df["Age Group"].fillna(
    df["Age Group"].mode()[0]
)

df["Hotel"] = df["Hotel"].fillna(df["Hotel"].mode()[0])

df["Food preference"] = df["Food preference"].fillna(
    df["Food preference"].mode()[0]
)

df["Rating(1-5)"] = df["Rating(1-5)"].fillna(
    df["Rating(1-5)"].median()
)

df["No of pax"] = pd.to_numeric(df["No of pax"], errors="coerce")
df["No of pax"] = df["No of pax"].fillna(
    df["No of pax"].median()
)

df["Bill"] = df["Bill"].fillna(df["Bill"].mean())

df["Estimated salary"] = df["Estimated salary"].fillna(
    df["Estimated salary"].median()
)

print(df)
```

```
   Customer ID Age Group  Rating(1-5)       Hotel Food preference
Bill  \
0            1     20-25          4.0         Taj             Veg
300.0
1            2     30-35          5.0         Taj         Non-veg
200.0
2            3     25-30          4.0  Lemon Tree             Veg
1322.0
3            4     20-25          4.0     Red Fox         Non-veg
1234.0
4            5       35+          3.0     Red Fox             Veg
989.0
5            6       35+          3.0         Taj         Non-veg
1905.0
6            7       25+          4.0         Taj             Veg
1000.0
7            8     20-25          4.0  Lemon Tree         Non-veg
```

```
2999.0
8            9     25-30          2.0  Lemon Tree              Veg
3456.0
9           10     30-35          5.0         Taj              Veg
3456.0
```

	No of pax	Estimated salary
0	2	45000.0
1	3	50000.0
2	2	120000.0
3	2	45000.0
4	2	122220.0
5	1	211234.0
6	1	345623.0
7	3	30000.0
8	3	99999.0
9	5	99999.0